

Algebraic Datatypes in Isabelle/HOL

```
theory Algebraic_Datatypes
imports Main
begin
```

Algebraic data types are a fundamental way to model data in functional programming. The following examples show how we model natural numbers, lists, binary trees with data in the inner nodes, and binary trees with data in the leaves.

```
datatype Nat = Zero | Succ Nat
```

```
term "Zero"
term "Succ (Succ Zero)"
```

```
datatype 'a List = Nill | Conss 'a "'a List"
```

```
term "Nill"
term "Conss (2::nat) (Conss 3 Nill)"
```

```
datatype 'a tree = Leaf | Node "'a tree" 'a "'a tree"
```

```
term "Node (Node Leaf 2 Leaf) (1::nat) (Node Leaf 3 Leaf)"
```

```
datatype 'a tree' = Leaf' 'a | Node' "'a tree'" "'a tree'"
```

```
term "Node' (Node' (Leaf' (2::nat)) (Leaf' 1)) (Leaf' 3)"
```

```
end
```